The Universal OCR & Plate Detection System is architected to support two distinct delivery pathways tailored to the technical requirements and regulatory constraints of different customer personas:

• Pathway 1 (End-Users & General Public): Zero-install web delivery hosted via Streamlit Community Cloud. Users interact with the AI application through a secure, public HTTPS hyperlink on their browser without downloading any software or managing code.

Streamlit Community Cloud allows free, continuous cloud hosting directly synchronized with a GitHub repository. Follow these exact steps to deploy or re-deploy the application:

Step 2.1: Repository Preparation on GitHub

The project repository must be public and contain the following essential files at the root level:

- **src/app.py: The main entry point containing the Streamlit web dashboard interface.**
- **requirements.txt: Defines all pinned Python dependencies with CPU PyTorch index for rapid build.**
- **packages.txt: Specifies low-level Linux Debian packages (libgl1, libglib2.0-0) required by OpenCV.**
- **models/model_manifest.json: Documents CRAFT and CRNN architecture specifications.**

Step 2.2: Launching on Streamlit Cloud

1. Navigate to https://share.streamlit.io and sign in with your GitHub account.

2. Click the 'Create app' or 'New app' button in the upper-right corner.

3. Configure the three deployment parameters:

- Repository: pankajatd/ocr_detection_with_7SDLC_MultiAgents

- Branch: main

- Main file path: src/app.py

For clients with sensitive document workflows (financial invoices, medical receipts, confidential car plate logs), the system provides single-command containerized execution:

Step 3.1: Building & Starting the Container

Execute from the project root folder:

```
docker-compose up -d --build
```

Step 3.2: Verifying Container Health

Execute:

```
docker ps
```

Confirm that 'ocr_detection_service' shows status 'Up' with port mapping 0.0.0.0:8501->8501/tcp.

Step 3.3: Accessing the Dashboard

Open any browser on the local network and navigate to: http://localhost:8501

Step 3.4: Stopping the Container

Execute:

Before formal project sign-off, verify each capability against the acceptance checklist:

• [x] Benchmark 1 (Vehicle Number Plate): AI extracts 'MH 12 AB 1234' with plate disambiguation.

• [x] Benchmark 2 (Store Receipt): AI extracts itemized prices and totals through shadows.

• [x] Benchmark 3 (Printed Document): AI straightens 15° skewed document and extracts all 6 lines.

• [x] Benchmark 4 (Tilted Signboard): AI extracts caution text on 3D perspective angle.

• [x] Camera Ingestion: Live snapshot capture operates seamlessly via st.camera_input.

• [x] Multi-Format Export: 1-Click downloads generate clean TXT, CSV, and JSON files.

• [x] Automated Tests: 26/26 PyTest unit and integration tests passing in terminal.